

MICRO★CRAFT

THE SUMMER OF BUILDING



LOADING....



Microsoft Innovation Club

Wikirace.ai

Navigate knowledge. Faster than humans.

Python → Black Hole

Chess → Game Theory

Cricket → Neural Networks

Music → Fourier Transform

Elon Musk → SpaceX

Machine Learning → Google



by Anas Arfeen



The Human Game



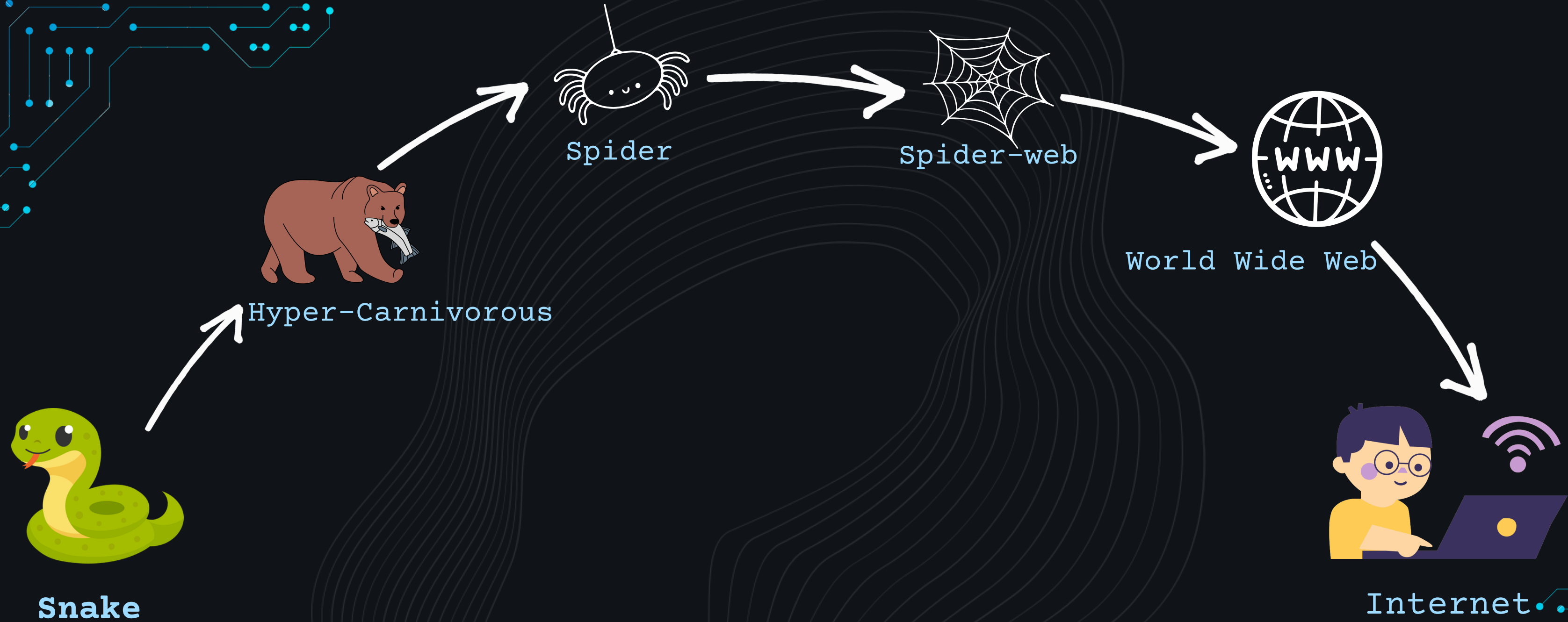
Snake



Internet

“What path should we take?”

The Human Game



"This is what I took 😱!"

The Human Game



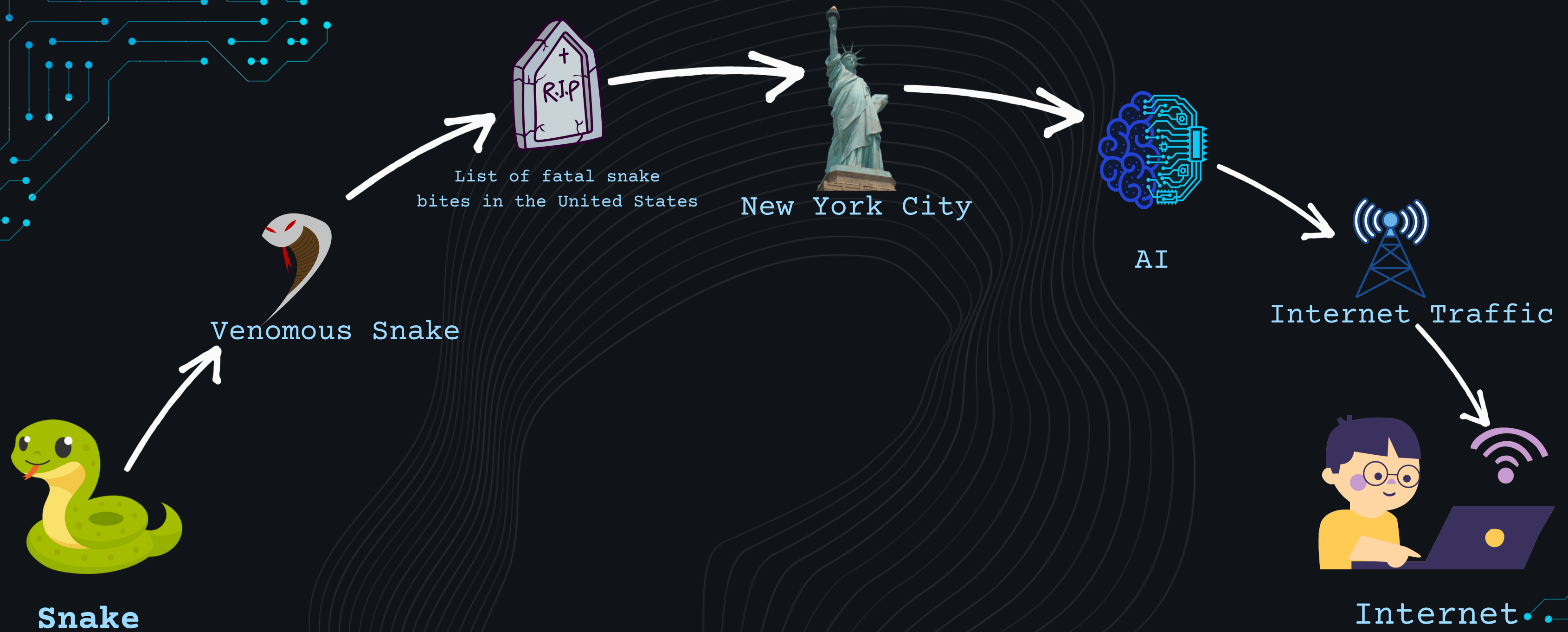
Snake



Internet

“What other path can we take?”

The Human Game



"This is what my friend took 🙌🙌!"

Humans Think Weirdly

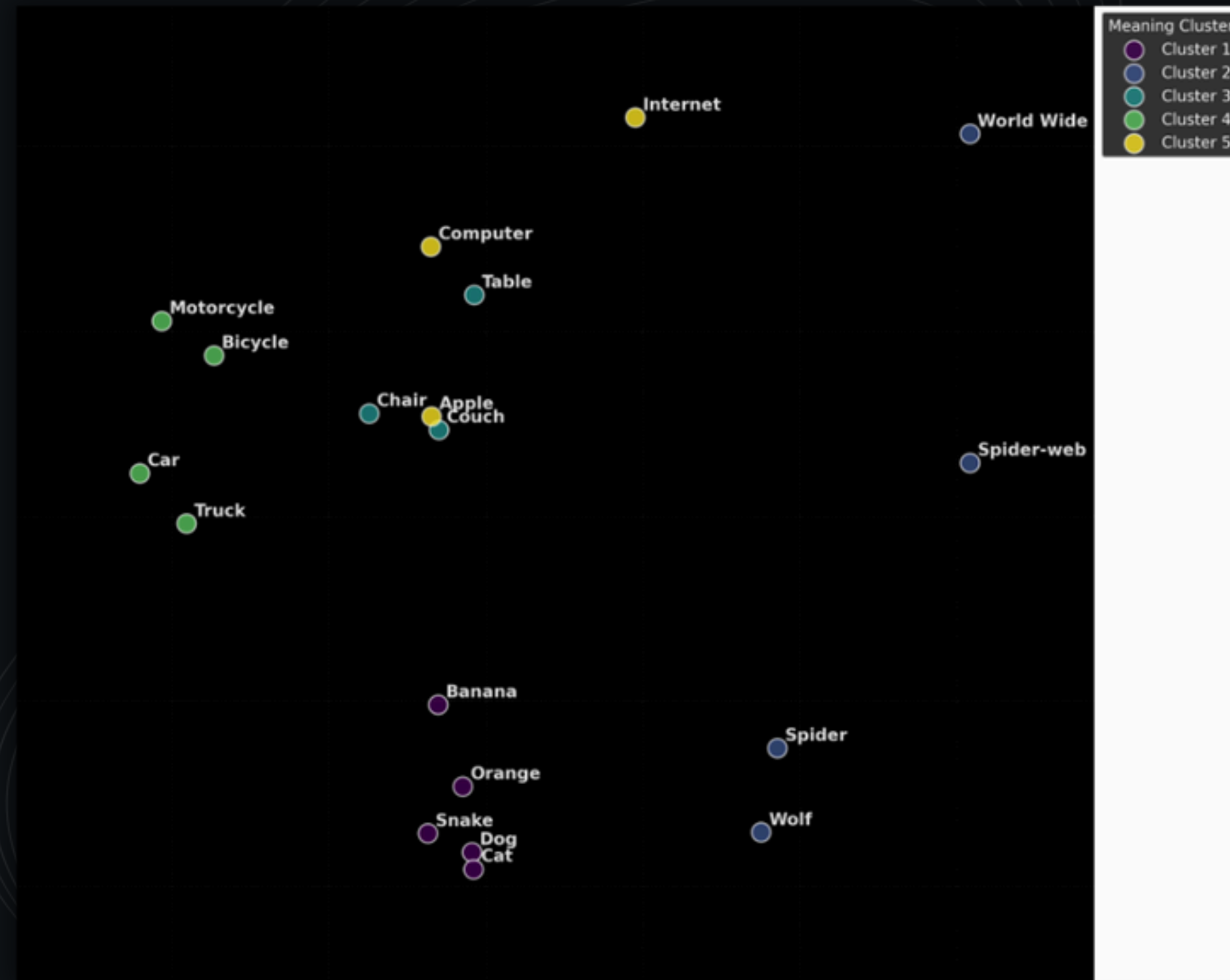
Mine

Snake → Hyper-Carnivorous → Spider → Web → WWW → Internet 🤔

My Friend's

Snake → Venomous snake → List of fatal snake bites in the United States → New York City → Artificial intelligence → Internet traffic → Internet

How does AI understand meaning?



Dog $\approx$ Cat

Embeddings

PICK A QUERY ITEM

cat

dog

kitten

puppy

lion

car

truck

bicycle

apple

banana

QUERY VECTOR

[0.70, 0.20, 0.10, 0.60] animal · pet · soft

PICK A QUERY ITEM

cat

dog

kitten

puppy

lion

car

truck

bicycle

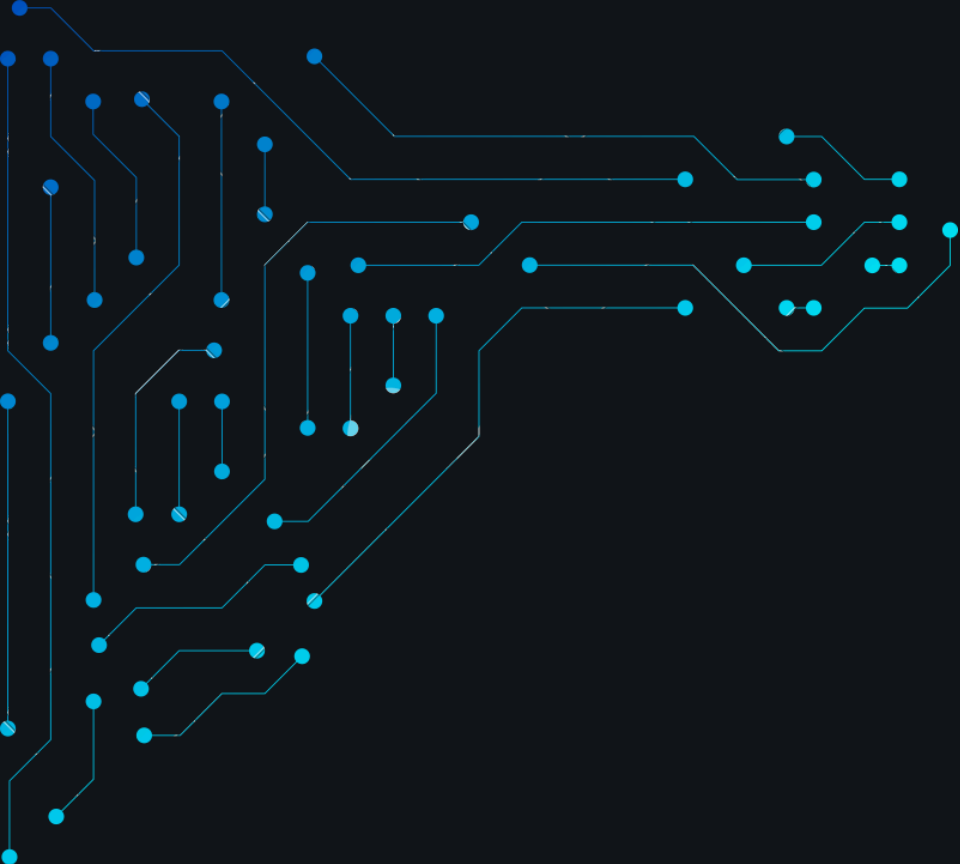
apple

banana

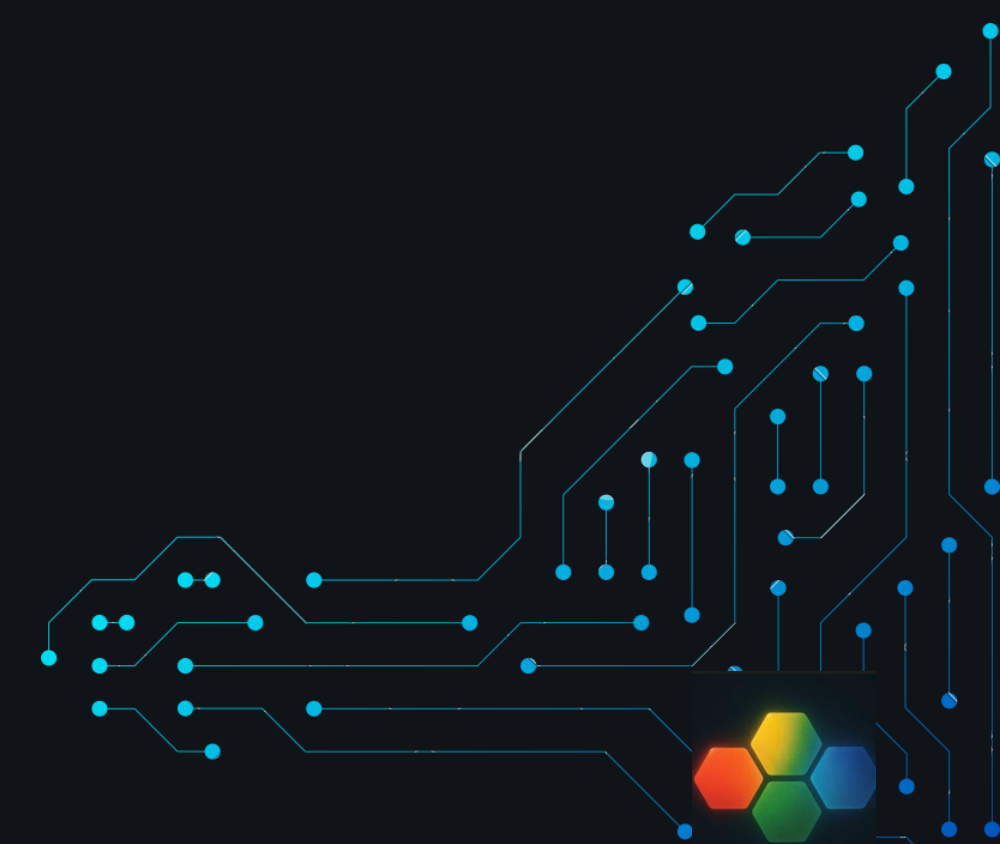
QUERY VECTOR

[0.70, 0.30, 0.10, 0.50] animal · pet · loud

These coordinates are called embeddings.



Be Right Back



Bot Psychology: Meet the Bots



Greedy Bot

Always picks the highest score



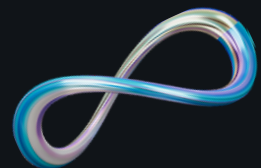
Explorer Bot

Willing to score lower to go further



Random Bot

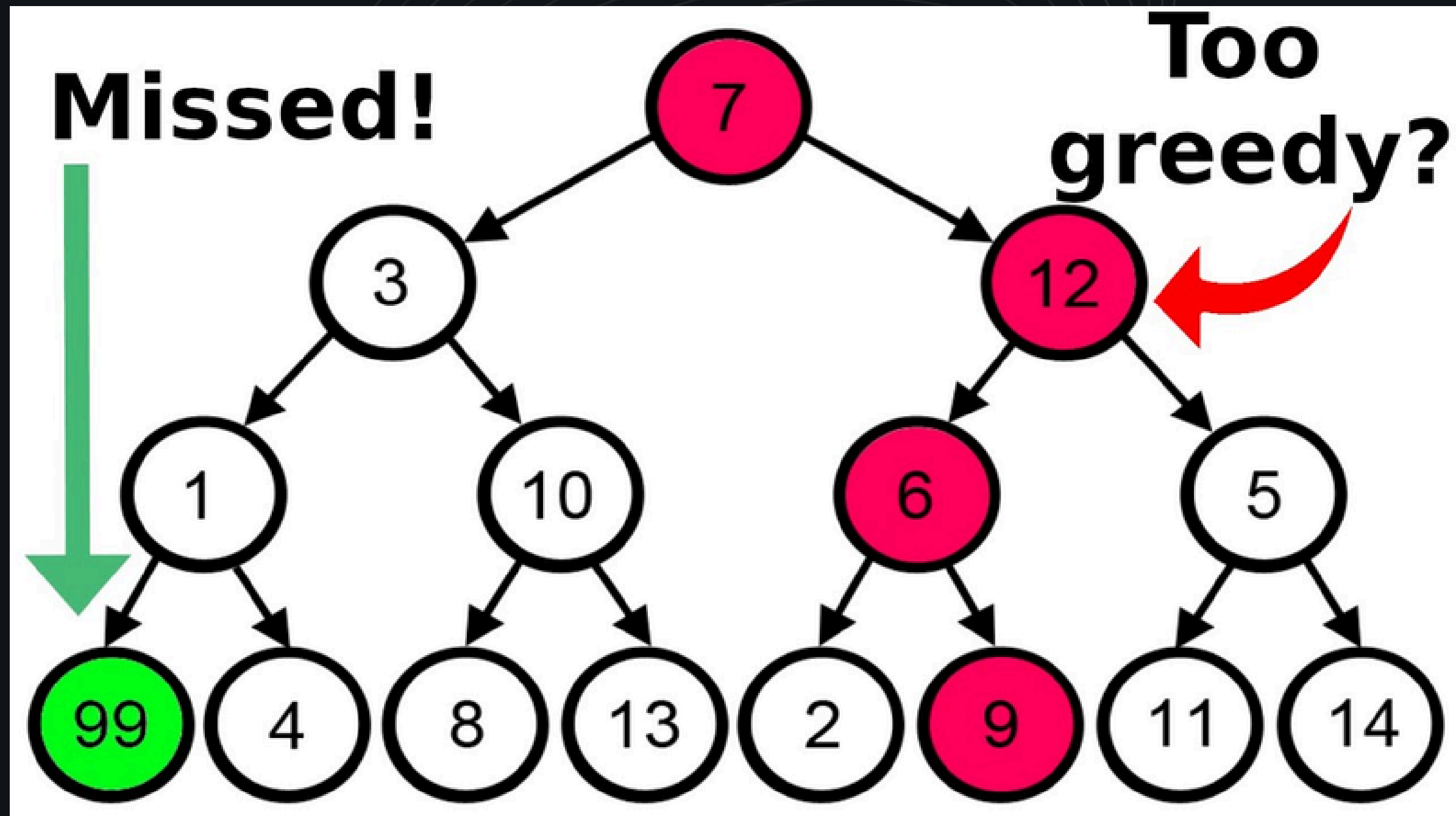
Chooses randomly.



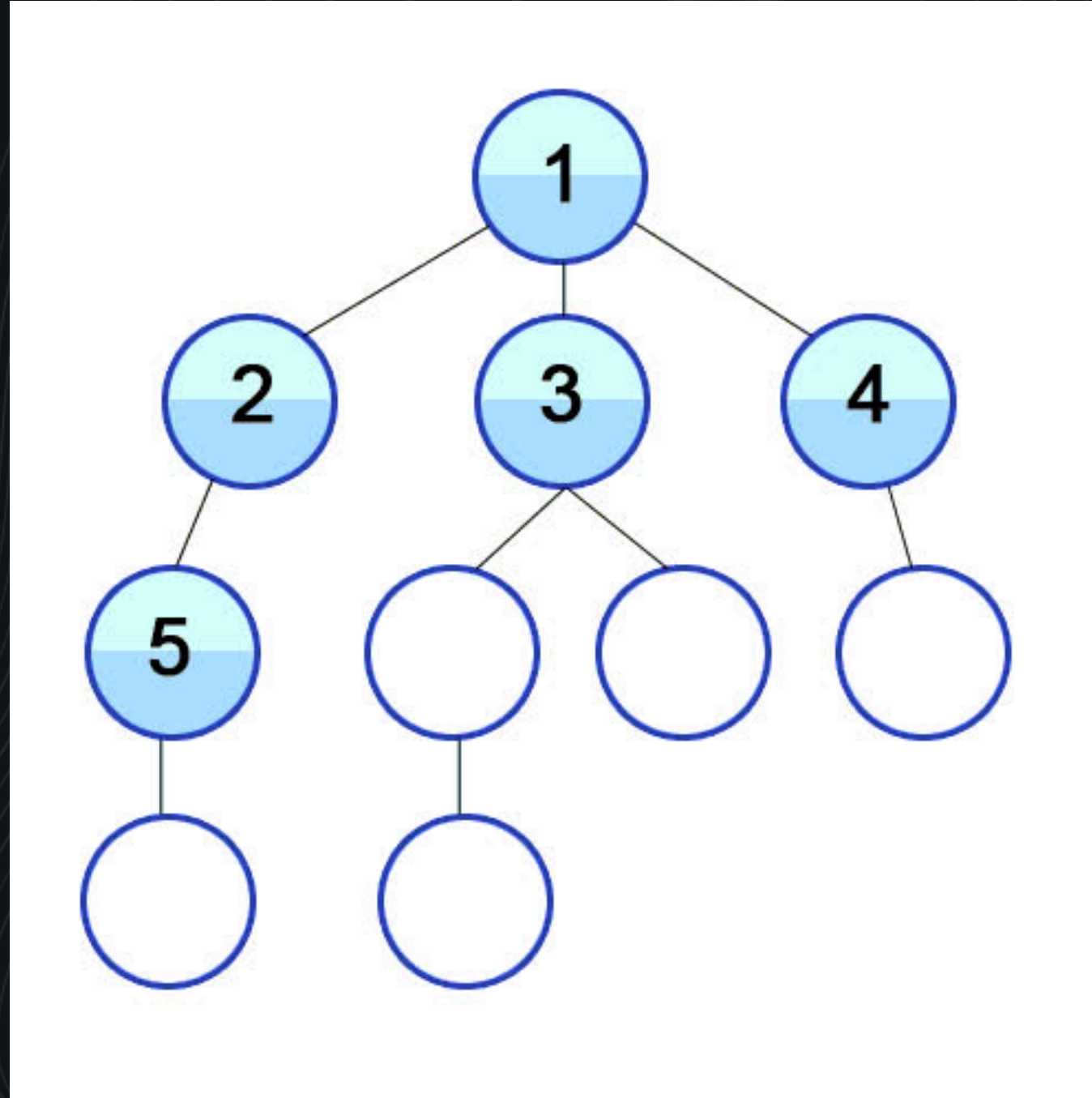
Looping Bot

Gets trapped. Can't escape

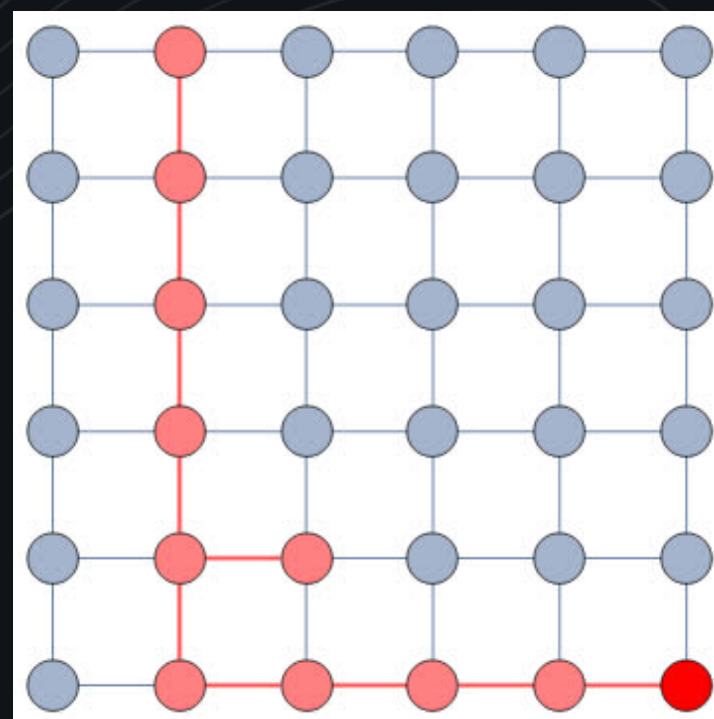
Greedy Bot



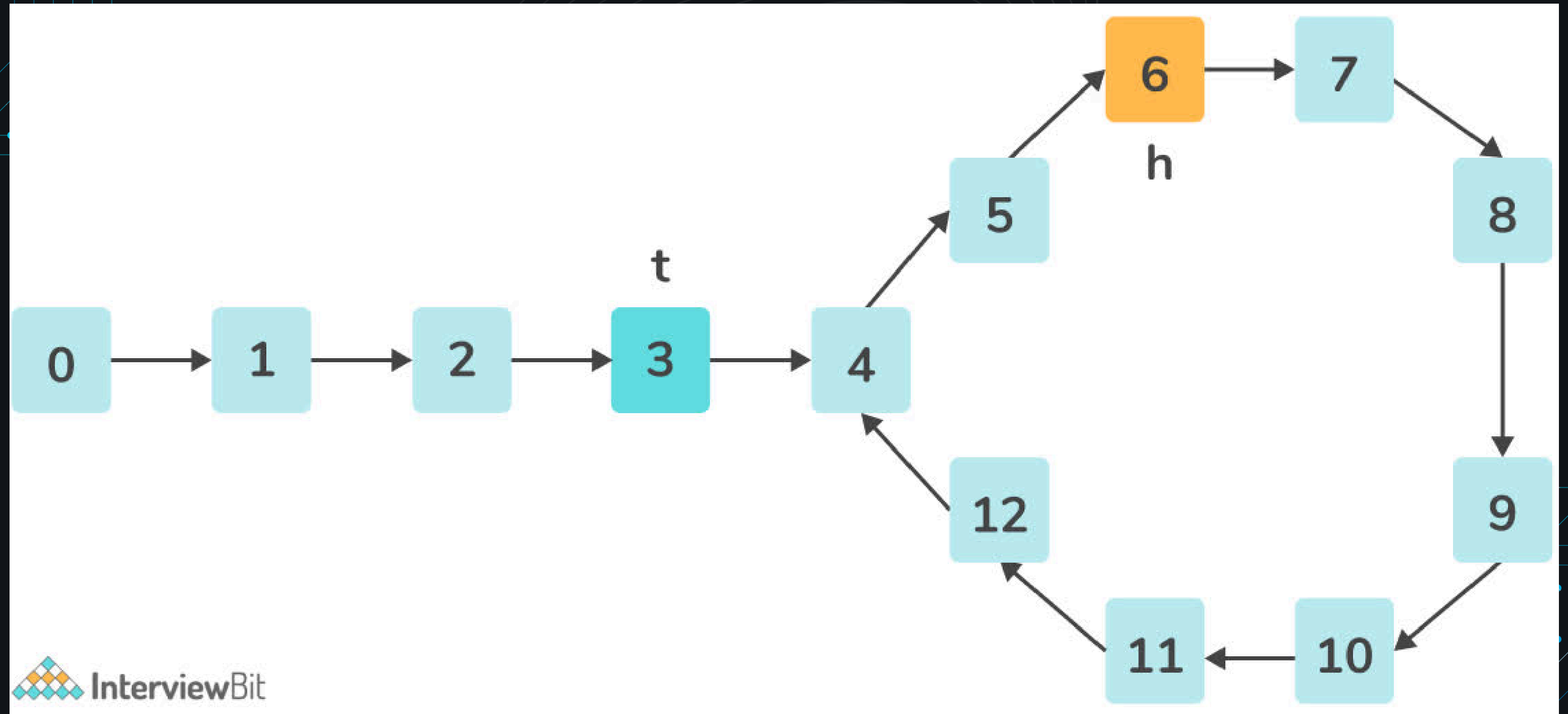
Explorer Bot



Random Bot



Looping Bot



Exploration vs Exploitation

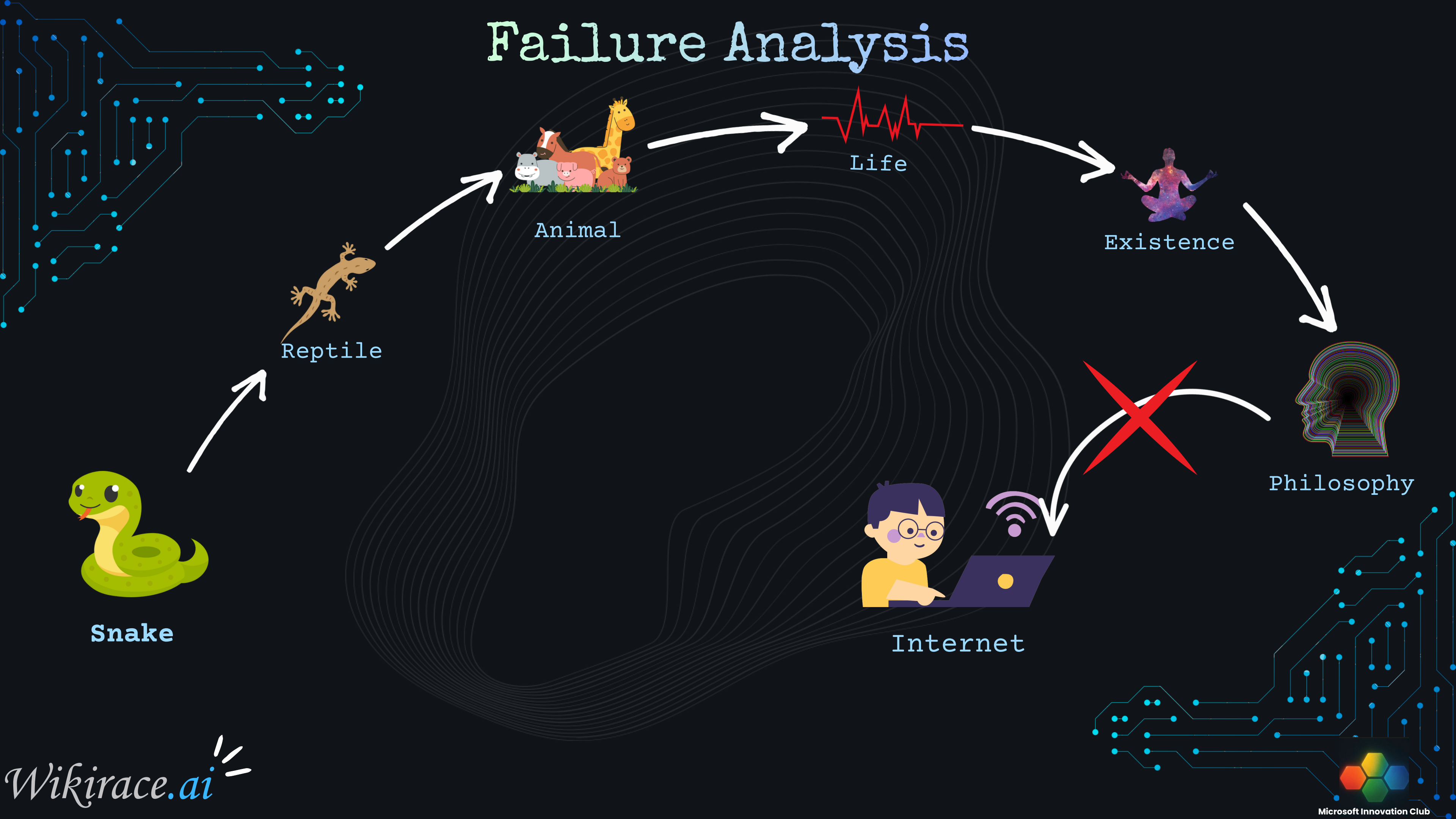
One of the most fundamental concepts in all of AI

Exploration



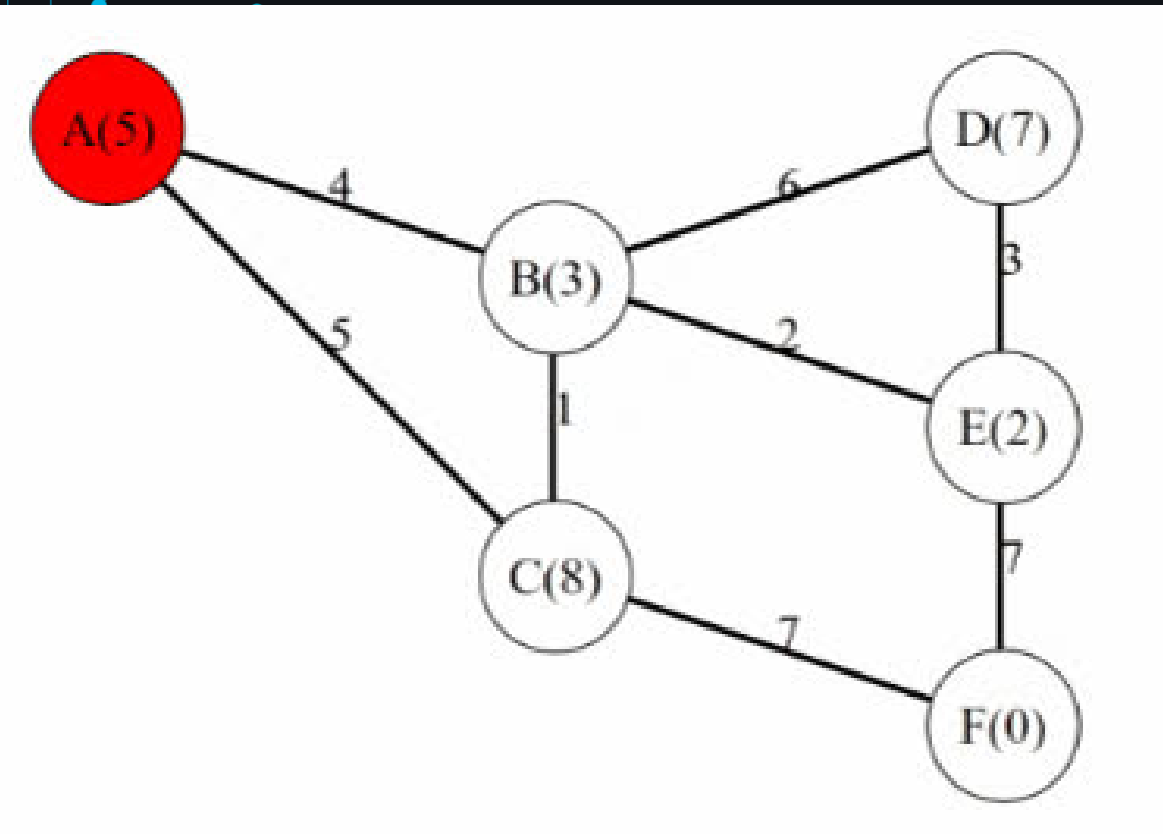
Exploitation

Failure Analysis

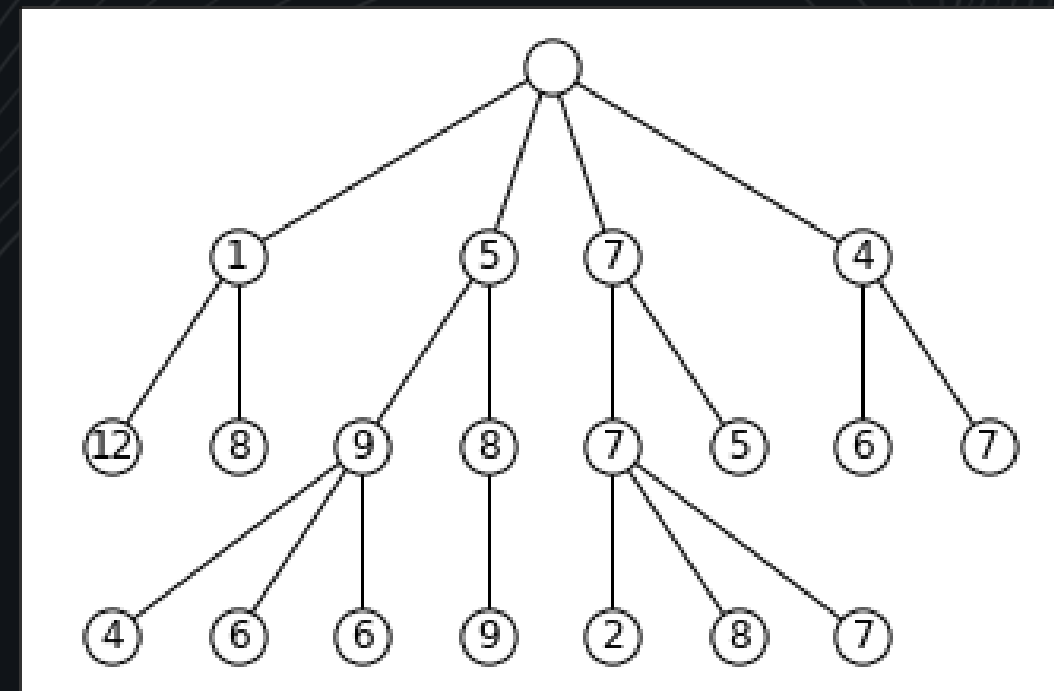


Search Algorithms

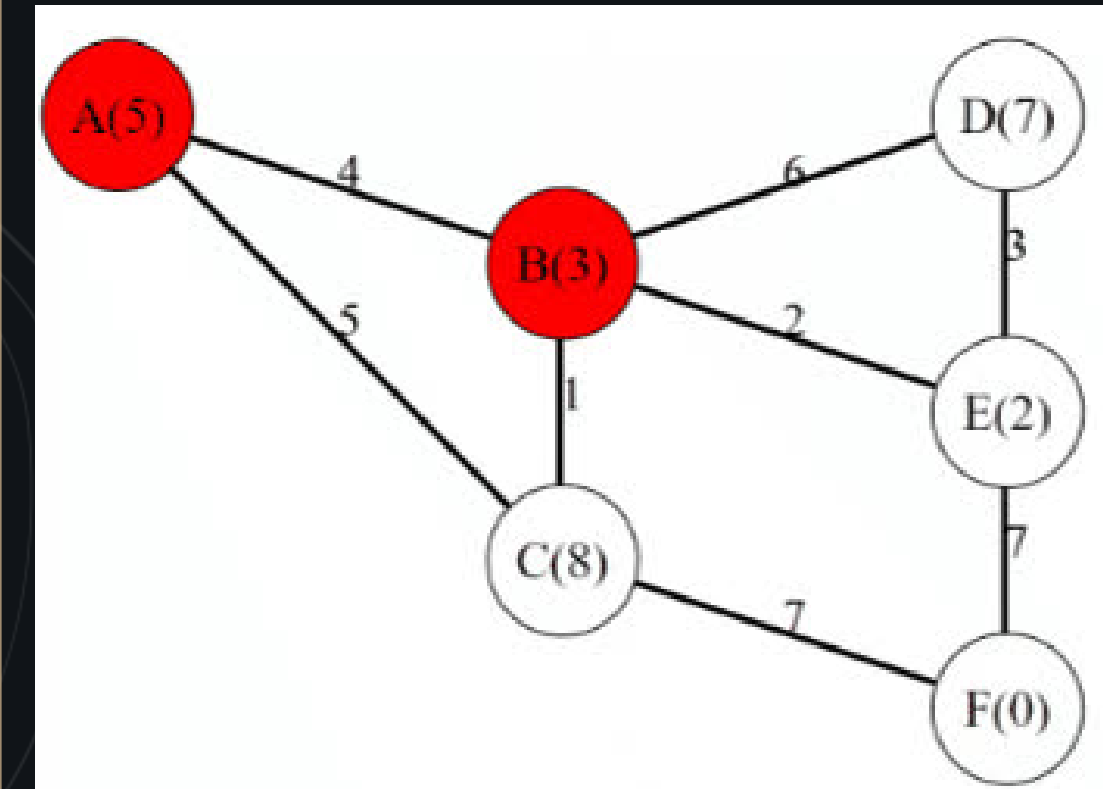
Greedy Search



Beam Search



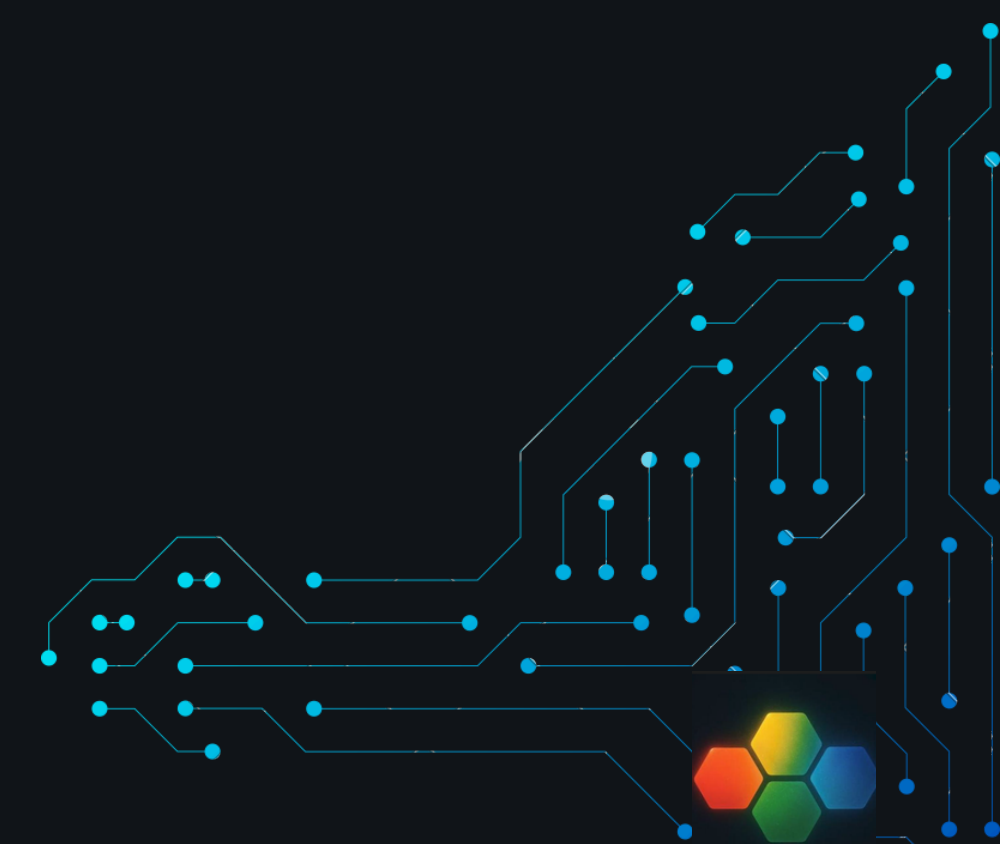
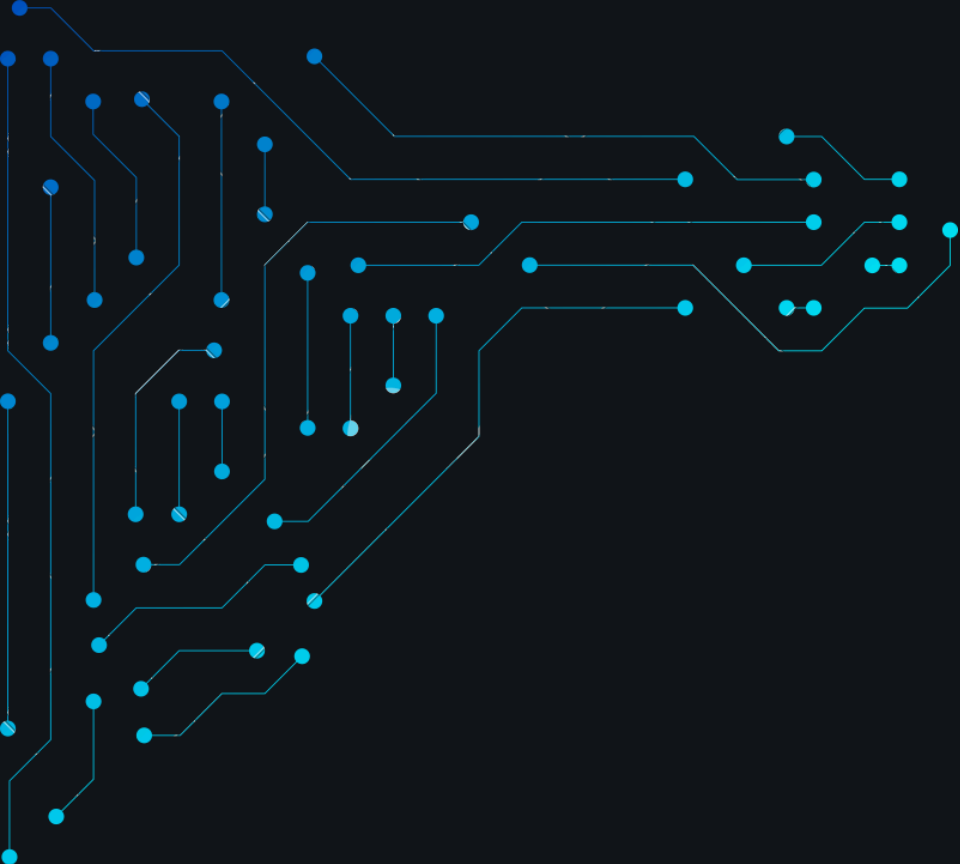
A* Search



Attention: The “it” Exercise

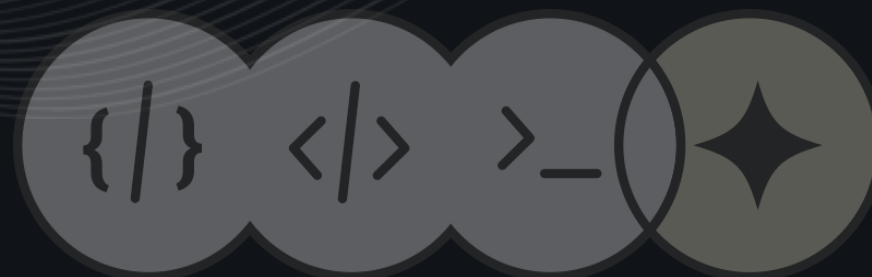
“The animal didn’t fit in the suitcase because it was too big.”

“The suitcase didn’t fit in the car because it was too big.”

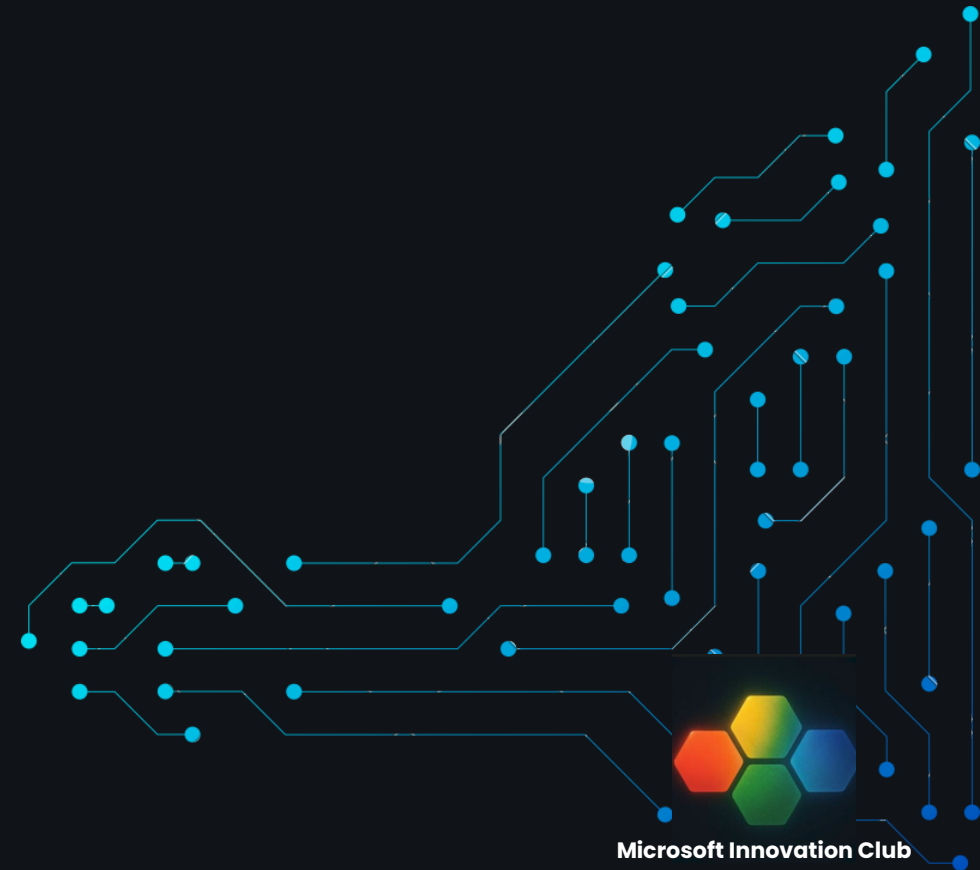


Be Right Back

IT'S CODING TIME



Thank You



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